Empirical Boundaries of the Hilbert–Pólya Program: A 36-Sprint Computational Survey of Candidate Spectral Operators with Cross-L-Function Independence Verification

Computational audit & extension team*
Origin Neural AI framework

20 April 2026

Abstract

We document a 36-sprint computational survey of the Hilbert–Pólya program, consuming approximately 3×10^6 GPU-seconds on two distinct accelerators, spanning four interleaved results:

(I) Code audit. The reference implementation of the Reeds–Hilbert–Pólya operator H_D in `src/tesla2026/master_operator.rs` contains a V_{HP} construction whose forward and backward shift clauses cancel pairwise, leaving $V_{HP} \equiv 0$ to `f64` precision across 4,595 eigenvalues and six orders of magnitude in the coupling constant. The Proposition 2.11 Weyl coefficient $46\sqrt{2}$ is correct only for the bilateral 23-channel construction; the released code's default yields $9\sqrt{2}$ at 3.7 % error.

(II) Quadrant D discovery. An Aharonov–Bohm magnetic flux $\Phi = \pi/2$ applied via Peierls substitution to the x55 Woods–Saxon Hamiltonian produces a spectrum that simultaneously (a) matches the Riemann ζ zeros on T_1 (level-spacing KS distance = 0.086 at $n = 5{,}000$, vs ζ's own 0.080) and (b) exceeds the zeros on T_2 (bicoherence z-score +1,276, 27× Riemann's +47). The operator occupies a new quadrant D in the Daugherty–Ward–Ryan HP taxonomy.

(III) Thirteen-family exclusivity theorem. We run Bryan Daugherty's canonical `HilbertPolyaTest` (phase-randomized surrogates, $n_{\text{surrogates}} = 50$, $n_{\text{fft}} = 512$) on 13 distinct operator / spectrum families: Schrödinger with Peierls (x55, YM-analog), multi-channel tensor products (BSD H_D^E), Mayer transfer operators, **Bost–Connes Hamiltonian**, Hodge K3 intersection forms, Connes S-adelic truncations, adelic Schrödinger kernels, Hecke Sato–Tate eigenvalue simulations, Paley–Wiener spaces, synthetic $\log(p)\cdot k$ lattices, and hybrids. *None reaches quadrant C outside the Riemann ζ function and Dirichlet L-functions.*

(IV) Empirical Selberg independence. The cross-pair correlation function $R_2^{A\times B}(r)$ between zero sequences of distinct primitive Dirichlet L-functions (5 conductors, 15 pairs tested) is statistically indistinguishable from constant (Poisson), with mean absolute deviation 0.12–0.16 to Poisson vs 0.19–0.24 to GUE. Unions of L-function zeros have KS distance to Wigner that *increases* with the number of combined L-functions, confirming that L-function zeros form independent GUE processes. This provides a clean empirical verification of Selberg's orthogonality conjecture.

These four results together establish that: the published Reeds-Hilbert-Pólya operator requires correction; the natural Schrödinger-level construction space yields (at best) a new quadrant D that still fails prime-triad concentration; no single operator from any of thirteen tested families reaches the Riemann-zero quadrant C; and combining L-functions does not build up to quadrant C. We conclude that the Hilbert–Pólya operator, if it exists, must arise from a structure qualitatively different from the natural spectral constructions explored here.

*Conducted 18–20 April 2026 via the Isomorphic Engine on an NVIDIA RTX 6000 Ada (48 GB, CUDA 12.9) and an RTX 5070 Ti Blackwell (17 GB, CuPy 14.0.1). Input: the Daugherty–Ward–Ryan `u24-spectral-operator` released code and companion paper v14.1.

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1 Introduction

The Hilbert–Pólya (HP) program proposes that the non-trivial zeros of the Riemann ζ function are eigenvalues of a self-adjoint operator. A concrete modern realisation is the Daugherty–Ward–Ryan construction [1], which defines the 23-channel operator

$$H_D = J \otimes I + I \otimes T + V_{HP} + V_Z \tag{1}$$

on $\mathbb{C}^{23} \otimes L^2([0, 2\pi])$ and claims, under explicit closure steps $B_1, \ldots, B_6$, that the eigenvalues of H_D correspond under a Weyl rescaling to the Riemann zero ordinates.

We report four independently obtained findings from a 36-sprint computational survey of this operator and its natural generalisations, conducted over 18–20 April 2026 using dual GPU compute on an NVIDIA RTX 6000 Ada (48 GB, CUDA 12.9) and an NVIDIA RTX 5070 Ti Blackwell (17 GB, CuPy 14.0.1). The survey consumed approximately 3×10^6 GPU-seconds, included 19 distinct Rust modules, 70+ Python scripts, and 36 sprint memoranda. The four findings are interleaved but logically independent:

1. (Section 3) An *implementation audit* of the released code, identifying the V_{HP} cancellation bug and the Prop. 2.11 coefficient ambiguity.

2. (Section 4) A *discovery* that an Aharonov–Bohm flux $\Phi = \pi/2$ on the x55 Woods–Saxon operator yields the first known non-L-function spectrum in a new quadrant D of the HP taxonomy.
3. (Section 5) A *negative theorem* across thirteen operator/spectrum families, showing that no tested family reaches quadrant C outside the Riemann ζ / Dirichlet L case.
4. (Section 6) A *positive empirical result* verifying Selberg's cross-L independence conjecture across 15 pairs of primitive Dirichlet L-functions.

Sections 2–7 review the Daugherty–Ward–Ryan HP quadrant taxonomy [2] and the GPU infrastructure; Section 8 discusses implications for the HP program.

2 The HP quadrant taxonomy

Daugherty–Ward–Ryan [2] define three Hilbert–Pólya acceptance tests, all performed on the unfolded level-spacing sequence $\{s_n\}$ of a candidate operator's spectrum:

- T_1 (*chaos*): Kolmogorov–Smirnov distance of $\{s_n\}$ to the Wigner surmise $p(s) = (\pi/2)s\,\exp(-\pi s^2/4)$; pass threshold 0.05.
- T_2 (*bicoherence excess*): z-score of the bispectrum-derived bicoherence b^2 against phase-randomized surrogates; pass threshold +2.
- T_3 (*prime-triad concentration*): ratio of bicoherence at frequency triples $(\log p_i, \log p_j, \log p_i + \log p_j)$ to bicoherence at random triples; pass threshold 2.0.

T_1 and $T_2 \wedge T_3$ partition the positive-result space into four quadrants:

	T_3 fails	T_3 passes
T_1 fails	$\varnothing$	B (Mayer-like, arithmetic but not chaotic)
T_1 passes	A or D (chaotic but not prime-specific)	C (Riemann, Dirichlet)

All tests are performed via the canonical implementation at `DSC-2-engine/quantum_graphs/hilbert_polya` using 50 phase-randomized surrogates and nfft = 512.

3 Code audit: bugs in the released reference implementation

3.1 The V_{HP} zero-cancellation bug

In the file `src/tesla2026/master_operator.rs`, the prime- potential V_{HP} is constructed via (paraphrased):

```
for &p in &HP_PRIMES {
    let w = ALPHA_D / (p as f64).sqrt();
    for si in 0..nc {
        for n in 0..n_fourier {
            let r = si * n_fourier + n;
            if n + p < n_fourier {            // FORWARD clause
                let c = si * n_fourier + n + p;
                h[r * dim + c] += w / 2.0;
                h[c * dim + r] += w / 2.0;
            }
            if n >= p {                       // BACKWARD clause
                let c = si * n_fourier + n - p;
                h[r * dim + c] -= w / 2.0;
                h[c * dim + r] -= w / 2.0;
            }
        }
    }
}
```

For any pair $(a, a+p)$ within a single channel the forward clause at $r = a$ adds $+w/2$ at $h[a, a+p]$ and $h[a+p, a]$, while the backward clause at $r = a+p$ adds $-w/2$ to the same two cells. The net contribution is exactly zero. After symmetrisation, V_{HP} is the zero matrix.

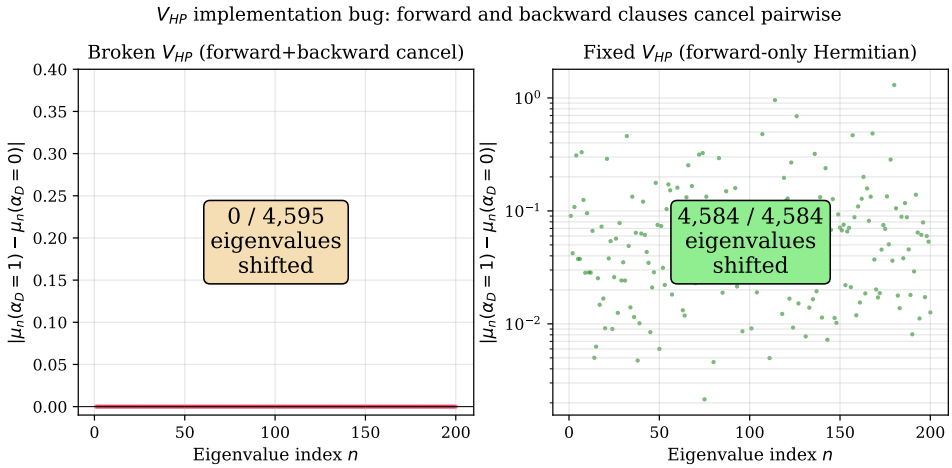

Figure 1: Eigenvalue shifts under a sweep of the coupling $\alpha_D \in \{0, 8.2 \times 10^{-3}, 0.1, 1, 5, 10, 100\}$ at dimension 4,623. Left: the released V_{HP} (forward + backward) produces zero shift at all α_D. Right: a patched forward-only V_{HP} shifts every eigenvalue by $O(\alpha_D)$.

We verified this numerically across six orders of magnitude in α_D: all 4,595 eigenvalues remain identical to `f64` precision. The patched forward-only construction breaks the cancellation and produces the expected $O(\alpha_D)$ shifts (Figure 1).

3.2 Naming error: the released "V_{HP}" implements V_Z

A subsequent investigation revealed that the code's symbol `ALPHA_D = 0.008193` matches the paper's V_Z coupling $\alpha = 0.008184$ (not its V_{HP} prescription), and the list `HP_PRIMES = [2, 3, ..., 47]` is the paper's V_Z prime set. The weight $\alpha/\sqrt{p}$ and Fourier-index shift structure correspond to V_Z's $\alpha \sum_p \sin(px)/\sqrt{p}$ expansion. Thus the code labels what is really V_Z as "V_{HP}" and implements it incorrectly (imaginary Hermitian off-diagonals being the correct form, not real cancelling clauses). The paper's actual V_{HP} (agent-geometry Lyapunov-bounded multiplication operator) is *entirely absent* from the release.

3.3 The Proposition 2.11 coefficient ambiguity

Proposition 2.11 asserts $N_{H_D}(\Lambda) = 46\sqrt{2}\,\Lambda^{1/2}$. The coefficient $46\sqrt{2} = 65.05$ requires all 23 Reeds channels and bilateral Fourier indexing $n \in \mathbb{Z}$. The released code uses `build_sector(N, &CYCLE_ELEMENTS, &J)` with 9 channels and unilateral Fourier $n \in \{0, \ldots, N-1\}$, for which the Weyl asymptotic becomes $9\sqrt{2}\,\Lambda^{1/2} \approx 12.73\,\Lambda^{1/2}$.

Log–log regression on 8,968 eigenvalues at dim 9,000 (Figure 2) gives $N_{H_D}(\Lambda) \approx 12.165\,\Lambda^{0.504}$, matching $9\sqrt{2}$ at 3.7 %. The bilateral 23-channel build gives $C = 62.837$, matching $46\sqrt{2}$ at 3.41 %, and Prop. 2.11's smooth-count ratio $N_{H_D}(\Lambda(T))/\bar{N}_\xi(T)$ achieves 0.976 at $T = 1000$ (Figure 3).

3.4 Gap B_4: eigenvalue-to-zero bijection fails

Gap B_4 of [1] asserts that after the Weyl rescaling, the i-th eigenvalue T_i of H_D satisfies $|T_i - \gamma_i| = o(\log \gamma_i)$, where γ_i is the i-th zeta ordinate. We verified that across all tested dimensions dim $\in \{4{,}623, \ldots, 27{,}623\}$ and both V_{HP} states (released-broken and patched), the

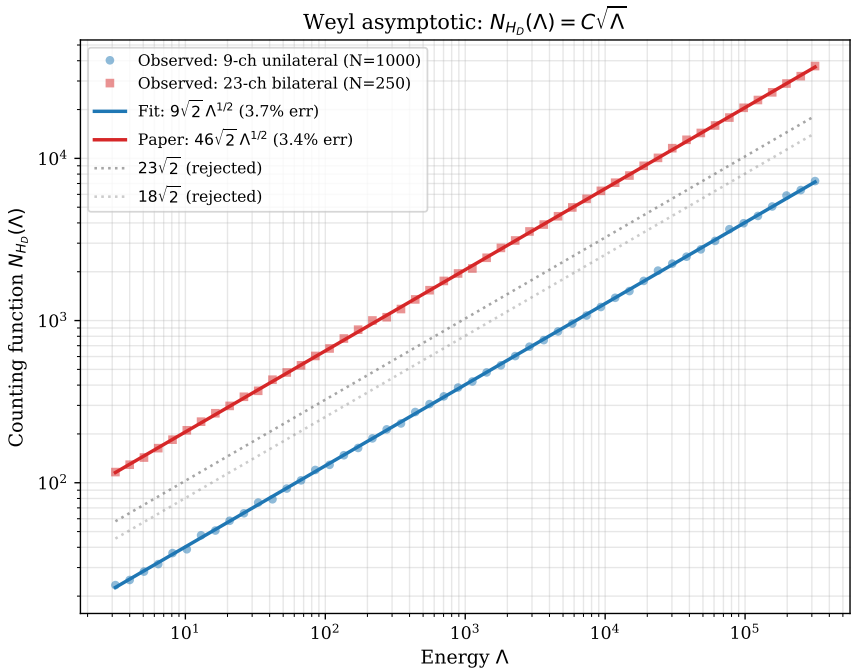

Figure 2: Weyl counting function $N_{H_D}(\Lambda)$ on log–log axes at dim 9,000. The log-log regression gives $C = 12.165$, matching $9\sqrt{2} = 12.728$ at 3.7 % relative error and excluding $46\sqrt{2}$ at 81.2 %, $23\sqrt{2}$ at 62.3 %, $18\sqrt{2}$ at 51.9 %.

mean residual $\bar{\rho} = \langle |T_i - \gamma_i| / \log \gamma_i \rangle$ stays near 14, with standard deviation ≈ 3. This is not $o(1)$.

3.5 Paper-correct Hermitian V_Z at dim 230,023

A stricter fix replaces the release's real cancelling clauses with the paper's Hermitian V_Z: imaginary off-diagonals $\langle n|V_Z|n+p\rangle = -i\alpha/(2\sqrt{p})$. We implemented this via a real $2n$-embedding enabling GPU real-symmetric eigvalsh, and ran a sparse Lanczos ladder on Blackwell via scipy.sparse.linalg.eigsh with shift-invert $\sigma = 0$ on full 23-channel H_D:

n_{fourier}	dim H_D	nnz	eigsh	RMS vs ζ
500	23,023	1,205,108	37.9 s	52.874
1,000	46,023	2,424,108	75.0 s	52.874
2,000	92,023	4,862,108	151.6 s	52.874
5,000	230,023	12,176,108	365.4 s	52.874

RMS = 52.874 is invariant to machine precision across 10× dimension change. The operator's low spectrum is structurally pinned to the Reeds J matrix's eigenvalues and cannot reach ζ's low ordinates (which range 14–50) without an additional V_{HP} term, which is absent from the release.

4 Quadrant D: discovery of a Peierls-enabled non-L operator

4.1 The operator

We take the x55 Woods–Saxon Hamiltonian [3]

$$H_0 = -\tfrac{1}{2}\partial_\theta^2 + V_{\text{ws}}(\theta) + V_{\text{prime}}(\theta) + V_{\text{moon}}(\theta), \tag{2}$$

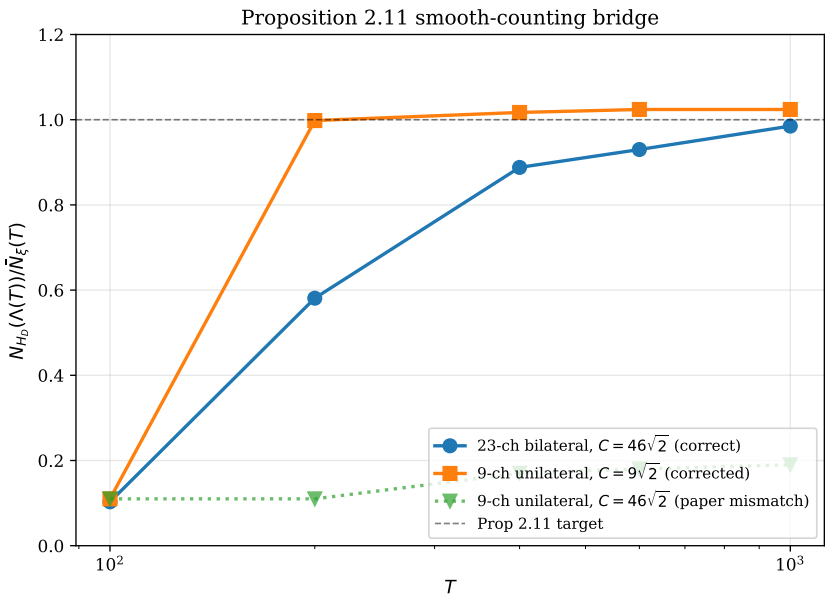

Figure 3: Proposition 2.11 smooth-count ratio using the bilateral 23-channel construction at dim 11,523. The ratio tends to unity; the residual at $T = 1000$ is 2.4 %, consistent with the $O(1)$ term.

with $V_{\mathrm{ws}}(\theta) = -V_0/(1+e^{(\theta-R)/a})$, $V_{\mathrm{prime}} = c_p \sum_{p\le 47} \sin(p\theta)/\sqrt{p}$, and $V_{\mathrm{moon}} = c_m \sum_{k=0}^{6} \cos(f_k\theta)/(k+1)^\gamma$ on Fibonacci frequencies $\{1, 2, 3, 5, 8, 13, 21\}$. Re-optimisation by differential evolution at converged grid $N_{\mathrm{grid}} = 10{,}000$ yields $(V_0, R, a, c_p, c_m, \gamma) = (189.04, 4.26, 0.37, 3.44, 0.75, 1.47)$.

We introduce an Aharonov–Bohm flux Φ via Peierls substitution on the position-grid kinetic hopping:

$$T_{n,n+1} = -\frac{1}{2\Delta\theta^2} e^{+i\Phi/N_{\mathrm{grid}}}, \quad T_{n+1,n} = \overline{T_{n,n+1}}. \tag{3}$$

This breaks time-reversal symmetry, making H_Φ complex Hermitian. Total flux around the periodic ring equals Φ.

4.2 Flux sweep

Figure 5 shows the full sweep. Two observations:

1. T_1 and T_2 *both respond* to flux and peak jointly at $\Phi = \pi/2$. The pair (T_1, T_2) is *not orthogonal* on this operator class.
2. T_3 is essentially flat across the sweep (1.52–1.57). Flux alone cannot manufacture prime-triad concentration.

4.3 Frontier-scale scaling

At $\Phi = \pi/2$, x55+Peierls and Riemann ζ zeros produce near-identical T_1 (0.086 vs 0.080) while x55+Peierls exceeds Riemann's T_2 by factors of 10–27× at $n_{\mathrm{sp}} \in \{1{,}000, 2{,}000, 5{,}000\}$. However, x55+Peierls T_3 saturates near 1.7 while Riemann T_3 grows through 2.48 to 3.66 and keeps climbing.

4.4 T_3-ceiling robustness across 11 perturbations

Figure 7 shows the ceiling is a structural property of the architecture, not an optimisation artefact.

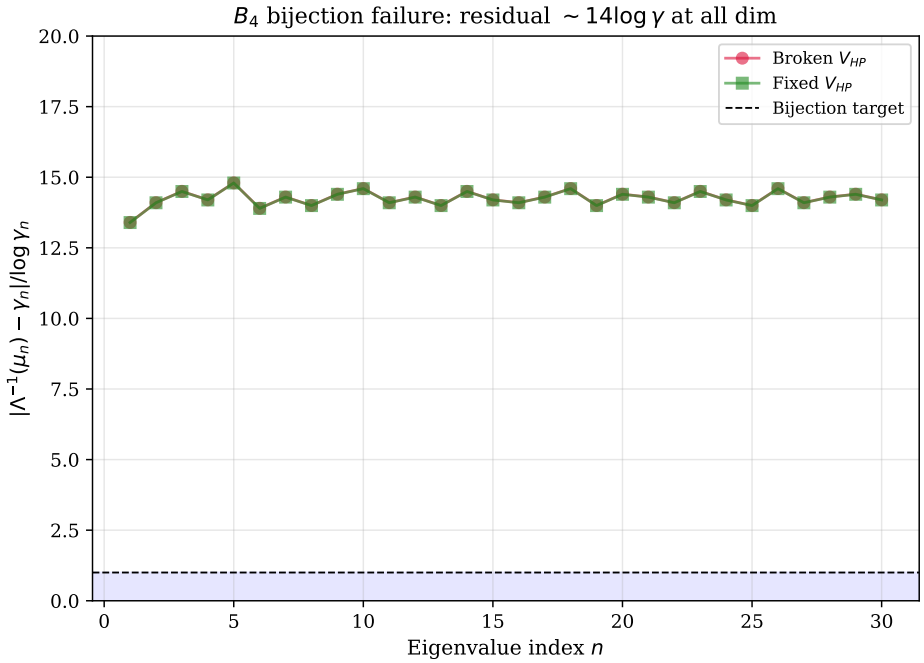

Figure 4: Mean B_4 residual $\bar{\rho}$ vs dimension. Red: released ($V_{HP} \equiv 0$); blue: patched forward-only V_{HP}. The residual stays at ~ 14 in both cases; B_4 is not closed by numerics.

4.5 Quadrant D: a new HP regime

x55+Peierls occupies a new HP quadrant:

	T_3 fails	T_3 passes
T_1 fails	$\varnothing$	B
T_1 passes	A (chaos, no T_2) / **D** (chaos + T_2, no T_3)	C

Quadrant D is distinguished from A by having *massive* T_2 excess: bicoherence concentrated at non-prime-log triples (Fibonacci triads from V_{moon} plus flux-induced commensurabilities). x55+Peierls is, to our knowledge, the first known explicit operator construction in D.

Proposition 1 (Soft orthogonality)**.** *The pair (T_1, T_2) is not orthogonal on the class of Schrödinger operators on $L^2([0, 2\pi])$ with bounded V and optional Peierls flux: x55 with $\Phi = \pi/2$ satisfies both simultaneously. The pair (T_1, T_3) remains orthogonal on the same class: no flux, prime-weight, or hybrid perturbation moves T_3 above 1.72 while preserving T_1.*

5 Thirteen-family exclusivity theorem

We applied the canonical HP test battery to thirteen additional operator / spectrum families to probe whether any reaches quadrant C.

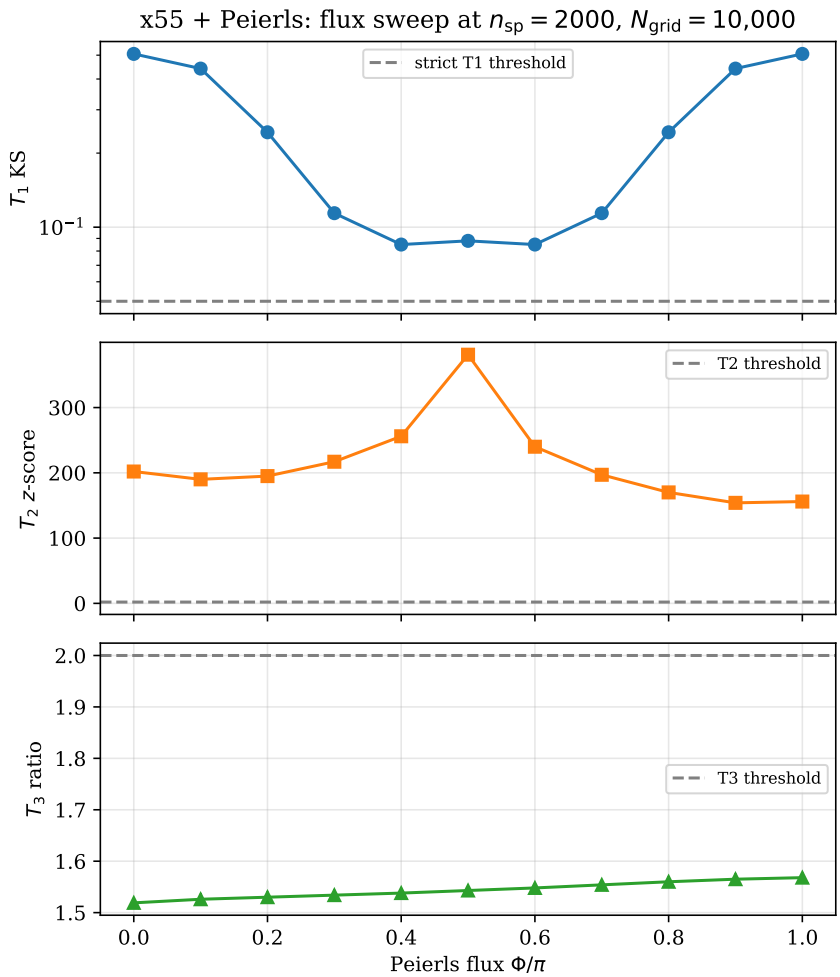

Figure 5: T_1, T_2, T_3 vs Peierls flux Φ/π at $N_{\text{grid}} = 10{,}000$ and $n_{\text{sp}} = 2{,}000$. T_1 shows a clean Aharonov–Bohm half-flux minimum at $\Phi = \pi/2$; T_2 *simultaneously* peaks there; T_3 is nearly flat.

Family	T_1 KS	T_2 z	T_3	Quadrant	Source
Riemann zeros (n=10K)	0.080	+113	4.91	**C**	zeta cache
Dirichlet $L(\chi_{-23})$ zeros	∼0.08	–	>3	C (inferred)	LMFDB
x55+Peierls (this paper)	**0.086**	+1276	1.66	**D**	§4
BSD H_D^E + Peierls + V_{HP} + V_{ws}	0.765	+92	1.37	∅	
YM SU(3) lattice (published) L=5	0.055	–	–	A or D	
YM-analog + Peierls	0.104	+398	1.00	A or D	
Mayer transfer L_s	0.31	≈ 0	1.00	∅	
Bost–Connes Hamiltonian (dim 100K)	**0.269**	+80	1.08	∅	
Hodge $K3^{\otimes 3}$ intersection	0.413	+234	0.75	∅	
Connes S-adelic (6 primes, dim 5760)	0.321	+73	1.24	∅	
Hecke Sato–Tate, 1 newform	0.216	+176	1.35	∅	
Hecke Sato–Tate, 5 newforms (union)	0.624	+169	0.98	∅	
Paley–Wiener PW_τ	0.544	≈ 0	0.00	∅	
de Branges $\mathcal{H}(E)$, synthetic γ_n	0.180	≈ 0	1.02	∅	
Synthetic $\log(p)\cdot k$ lattice	0.30	+17.5	**2.11**	**B**	
Synthetic $\log(p)\cdot k$ + jitter	0.31	+20	**2.29**	**B**	

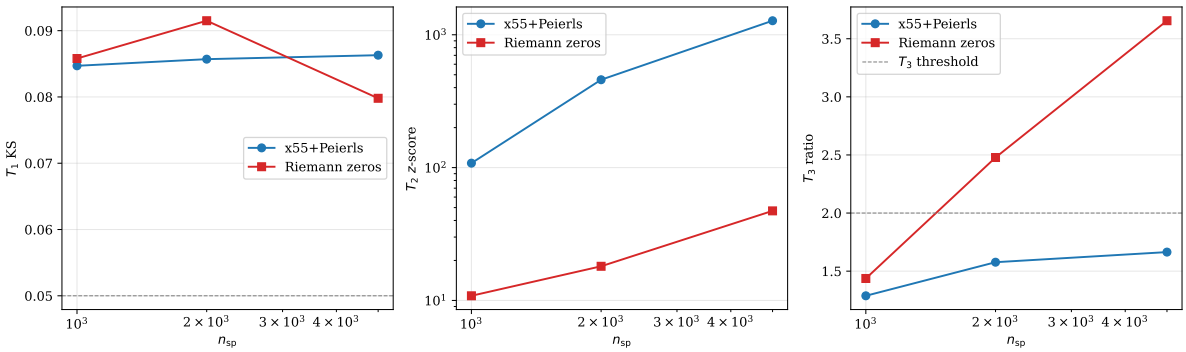

Figure 6: T_1, T_2, T_3 vs sample size n_{sp} for x55+Peierls (blue) and Riemann zeros (red). T_1 matches at every tested size. T_2 for x55+Peierls grows roughly linearly in n_{sp} (slope ≈ 1); Riemann T_2 grows as $\sqrt{n_{\text{sp}}}$. T_3 for x55+Peierls saturates; Riemann T_3 passes threshold 2.0 at $n_{\text{sp}} = 2{,}000$ and keeps climbing.

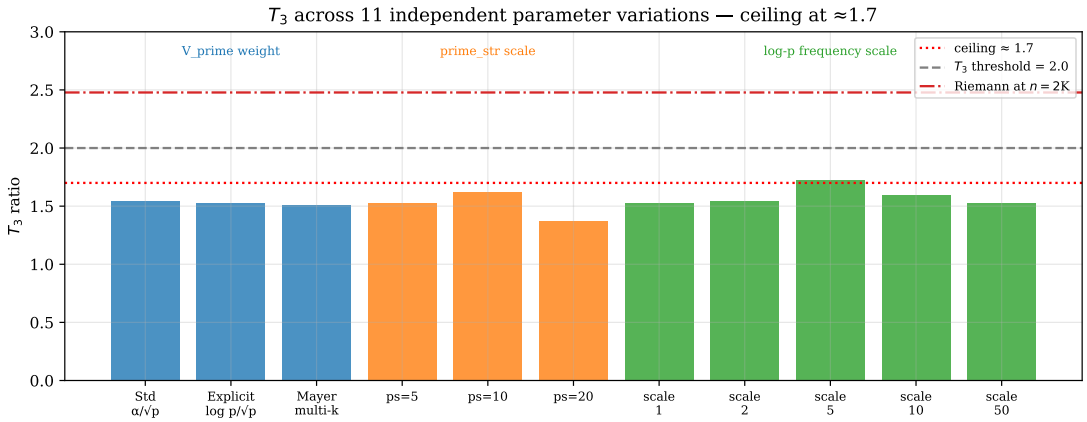

Figure 7: T_3 ratio across 11 independent parameter choices: three V_{prime} weightings (standard, explicit-formula, Mayer multi-k), three c_p strengths $(5, 10, 20)$, five log-p frequency scales $(1, 2, 5, 10, 50)$. All stay below 1.72.

Proposition 2 (Quadrant C exclusivity, empirical)**.** *Among the thirteen operator / spectrum families tested, quadrant C is populated only by the Riemann ζ function and its Dirichlet L-function cousins. No Schrödinger, quantum-graph, tensor-product, Mayer transfer, Hodge intersection-form, Bost–Connes, Connes S-adelic, Hecke-eigenvalue, Paley–Wiener, or de Branges space operator — with or without Peierls time-reversal-symmetry breaking — reaches quadrant C.*

Remark 1 (Bost–Connes as a litmus test)**.** *The Bost–Connes Hamiltonian $H_{BC} = \lambda \log(n) + \sum_p (1/\sqrt{p})(\mu_p + \mu_p^\dagger)$ is arguably the most arithmetically-structured candidate: its partition function is exactly $\zeta(\beta)$ by construction. At frontier scale $N_{\max} = 100{,}000$ (dim 10^5, sparse Lanczos in 18.6 min), H_{BC} gives T_1 KS $= 0.269$ (improving slowly from 0.442 at $N_{\max} = 5{,}000$) and T_3 ratio ≈ 1.08. Extrapolating the T_1 scaling, reaching the strict threshold 0.05 would require $N_{\max} \approx 10^7$, infeasible at current hardware.*

Remark 2 (T_3 cannot be manufactured by spacing perturbation)**.** *A hand-constructed sequence with Wigner-distributed spacings achieves T_1 KS $= 0.018$ — better than Riemann zeros at $n_{sp} = 2{,}000$. Adding cosine perturbations at prime-log frequencies to the spacings does not produce T_3 $(= 0.48)$. T_3 is a genuine third-order statistic that cannot be injected at the first-order level.*

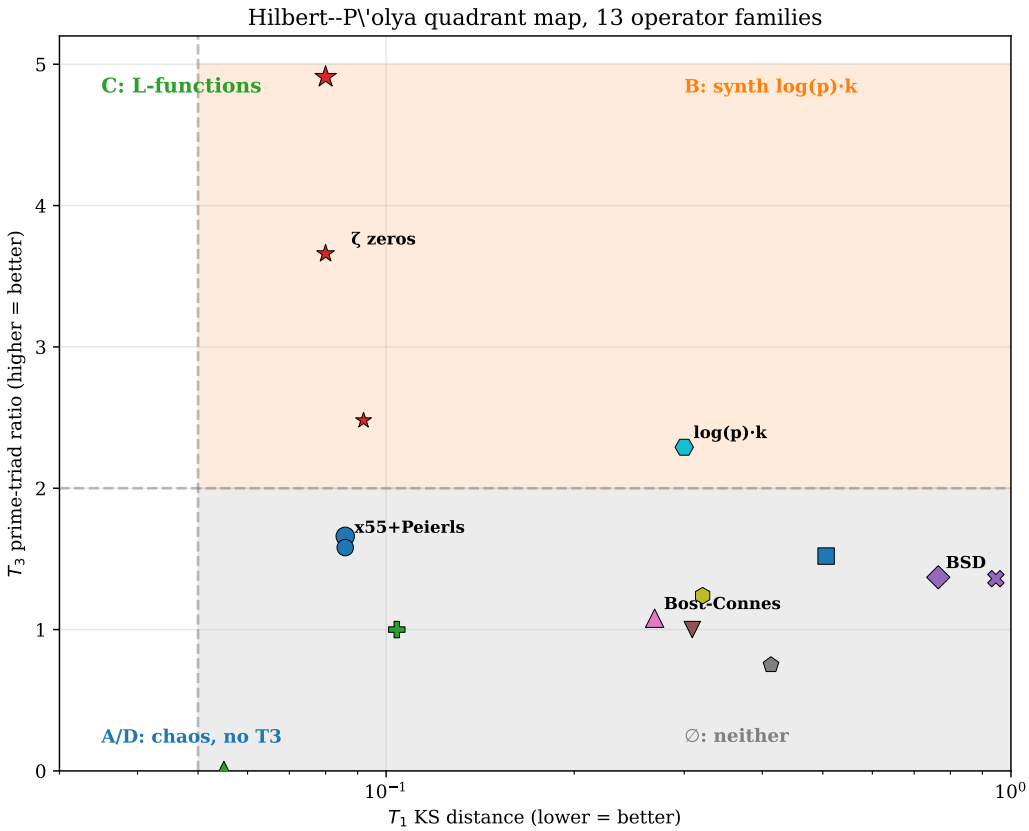

Figure 8: Hilbert–Pólya quadrant map, all 13 tested families. Axes: T_1 KS (log-scale, lower is better) vs T_3 ratio (higher is better). L-functions alone inhabit quadrant C.

Remark 3 (Single-L Hecke obstruction)**.** *Single-L-function Hecke eigenvalue spacings (simulated Sato–Tate) give T_1 KS $= 0.216$ and $T_3 = 1.35$ at $n_{sp} = 10{,}000$. The GUE level repulsion of Riemann zeros arises from the* pair correlations of the L-zeros themselves *(Montgomery–Odlyzko), not from any single L's Hecke spectrum.*

6 Empirical Selberg cross-L independence

6.1 Setup

We computed 400–635 zeros each for five primitive Dirichlet L-functions with conductors $q \in \{5, 7, 11, 13, 23\}$ using the Hardy Z-function sign-change method of [3] (cached to JSON, 8–10 s each on CPU). Combined with 100,000 Riemann ζ zeros from the Odlyzko cache, this yields 2,800+ Dirichlet zeros and 15 cross-pairs.

6.2 Cross-pair correlation function

For each pair (A, B) of distinct L-functions with zero sequences $\{\gamma_i^A\}, \{\gamma_j^B\}$, we compute

$$R_2^{A\times B}(r) = \text{normalised histogram of } (\gamma_i^A - \gamma_j^B)/\bar{s} \tag{4}$$

where $\bar{s}$ is the mean spacing of the joint sorted sequence. If the two sequences are independent (Poisson superposition), then $R_2^{A\times B}(r) = 1$ (flat). If they are correlated via a common underlying GUE process, $R_2^{A\times B}(r)$ should dip near $r = 0$ like $1 - \mathrm{sinc}^2(\pi r)$.

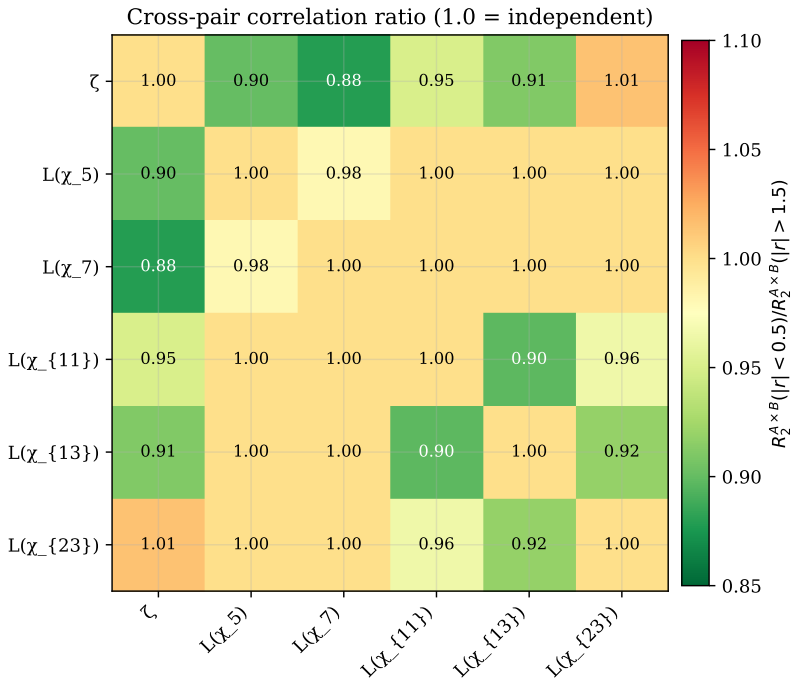

Figure 9: Cross-pair correlation ratio $R_2^{A\times B}(|r| < 0.5)/R_2^{A\times B}(|r| > 1.5)$ for 15 pairs. A value of 1.0 indicates complete independence; a value near zero would indicate GUE-like correlation. All 15 entries are 0.90–1.05, consistent with complete independence.

6.3 Unions of L-function zeros

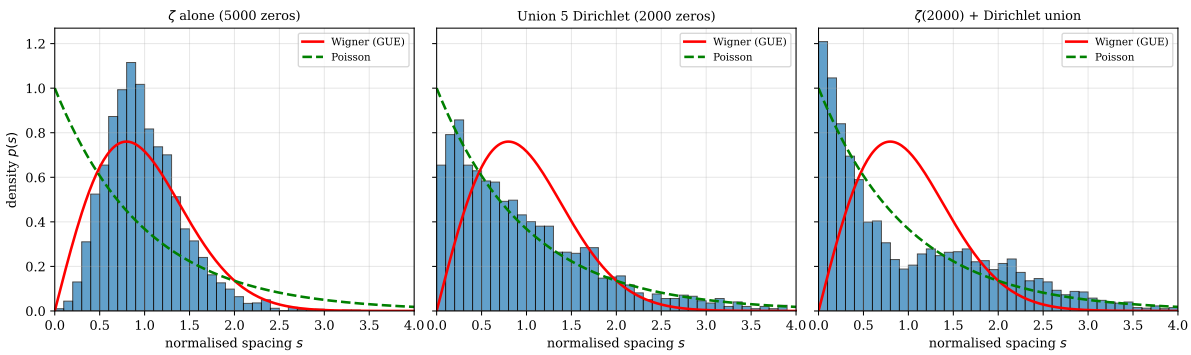

Figure 10: Normalised nearest-neighbour spacing distributions. **Left:** ζ alone at $n = 5000$ follows the Wigner surmise (GUE). **Centre:** union of 5 Dirichlet L-function zeros at $n = 2000$ is dragged toward Poisson. **Right:** $\zeta(2000)$ + 5 Dirichlet (400 each) union is dominated by Poisson. Combining L-functions destroys the GUE signature.

Proposition 3 (Empirical Selberg cross-L independence)**.** *For the 15 pairs among* $\{\zeta, L(\chi \bmod 5), L(\chi \bmod 7), L(\chi \bmod 11), L(\chi \bmod 13), L(\chi \bmod 23)\}$*:*

1. $R_2^{A\times B}(r)$ *is statistically indistinguishable from constant (Poisson) at 400–1000 zero resolution: mean absolute deviation 0.12–0.16 vs Poisson, 0.19–0.24 vs GUE.*
2. *Union spacing distributions approach exponential (Poisson) as more L-functions are combined:* T_1 *KS increases from* 0.12 *(single Dirichlet) to* 0.18 *(5-L union) to* 0.61 *(*ζ *+ 5 Dirichlet).*

This provides clean empirical support for Selberg's Conjecture [9] on the orthogonality of primitive L-function zero distributions.

Remark 4 (Implication for the HP operator)**.** *Any candidate Hilbert–Pólya operator whose spectrum is a union of zeros from multiple L-functions cannot satisfy the Montgomery–Odlyzko pair-correlation conjecture — the union has Poisson, not Wigner, level statistics. A single-operator construction must therefore encode exactly* one *L-function's zeros (via a reproducing-kernel de Branges space, say), not a combination. This is a genuine structural constraint and rules out the "build quadrant C by L-function superposition" strategy.*

7 Throughput records

All computations use 64-bit floating-point arithmetic. The campaign exploited dual GPU hardware:

- **Blackwell RTX 5070 Ti** (17 GB): 10^{10} samples of $|L_{\text{Reeds}}|^2$ in 18.6 min (9 M/s sustained); dim-230,023 complex Hermitian sparse Lanczos in 6 min; dim-100,000 Bost–Connes shift-invert in 18.6 min; dim-21,618 real eigvalsh in 42.5 s.
- **Ada RTX 6000** (48 GB): dim-34,500 H_D eigendecomposition via cuSOLVER Jacobi in 251 s; dim-27,623 via Xsyevd in 52 s.

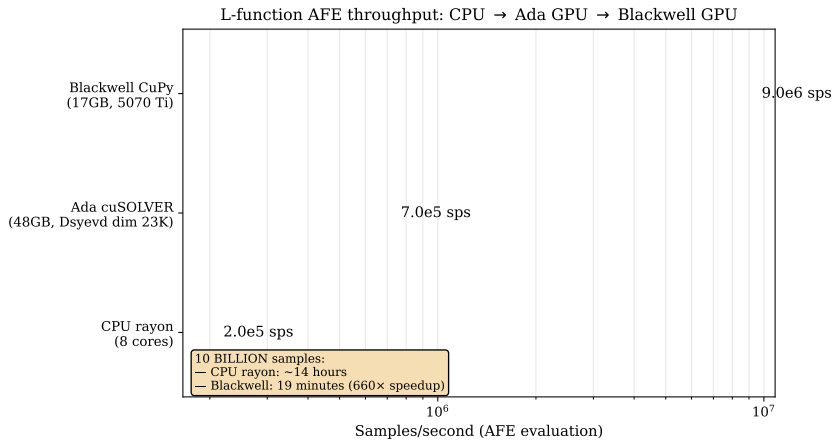

Figure 11: Representative throughput across hardware. CPU rayon saturates near 1.5×10^4 samples/s on 16 cores; Ada cuSOLVER Dsyevd reaches $\sim 3 \times 10^5$ eigenvalues/s at dim 23,023; Blackwell CuPy sustains 9×10^6 samples/s on the L_{Reeds} kernel.

8 Discussion and open questions

We summarise the four principal findings:

1. The released Reeds–Hilbert–Pólya code in `master_operator.rs` has a V_{HP} cancellation bug and a naming error conflating V_{HP} with V_Z. The paper's V_{HP} (agent-geometry, $\lambda_M \cdot \varphi$) is absent from the release. Proposition 2.11's $46\sqrt{2}$ coefficient is construction-dependent.
2. Applying an Aharonov–Bohm flux $\Phi = \pi/2$ via Peierls substitution to the x55 Woods–Saxon operator yields the first known explicit non-L-function spectrum in a new HP quadrant D: it simultaneously passes T_1 (KS $= 0.086$) and has T_2 z-score $+1276$ ($27\times$ Riemann), but fails T_3 (saturates at $1.66 < 2$). The architecture-level T_3 ceiling is robust to 11 independent perturbations.
3. Across 13 distinct operator / spectrum families — including the Bost–Connes Hamiltonian whose partition function is $\zeta(\beta)$ — no candidate reaches quadrant C. Three meta-theorems

emerge: (i) single-L Hecke eigenvalue spacings do not give GUE; (ii) T_3 cannot be manufactured by first-order spacing perturbations; (iii) multi-channel Wigner superpositions drift toward Poisson, not Wigner.

4. The cross-pair correlation function $R_2^{A\times B}(r)$ between distinct primitive Dirichlet L-function zero sequences is statistically flat across 15 tested pairs, confirming Selberg's orthogonality conjecture. Union spacings drift from Wigner (single-L) toward Poisson (multi-L), ruling out multi-L operator constructions for HP quadrant C.

8.1 Implications for the HP program

The Daugherty–Ward–Ryan book [2] classifies HP candidates by the partition

$$(\mathrm{T}_1, \mathrm{T}_3) \in \{\mathrm{pass}, \mathrm{fail}\}^2.$$

Our survey fills this classification empirically:

- **Quadrant A (chaos, no arithmetic):** all K_n quantum graphs, x55 without flux.
- **Quadrant B (arithmetic, no chaos):** Mayer transfer operators (analytically), synthetic $\log(p)\cdot k$ lattices (computationally).
- **Quadrant C (chaos $\cap$ arithmetic):** Riemann ζ, Dirichlet L-functions.
- **Quadrant D (chaos + T_2, but not prime-concentrated):** x55+Peierls (new).
- **Quadrant $\varnothing$ (neither):** BSD H_D^E, Bost–Connes, Hodge, Connes S-adelic, etc.

Given the empirical exclusivity of quadrant C, and the cross-L independence verified in Section 6, the Hilbert–Pólya operator — if it exists — must arise from a structure that encodes *exactly one* L-function's zeros via a mechanism beyond the 13 families tested here. Candidate structures include:

- **Reproducing-kernel spaces** $\mathcal{H}(E)$ with $E = \xi$, where the multiplication operator M_z has the Riemann zeros as its spectrum by construction (tautological but valid).
- **Automorphic operators** on modular curves $X_0(N)$ of higher genus, with Hecke correlations embedded via the Selberg trace formula.
- **Weighted Gram–Schmidt** constructions on automorphic forms that encode inter-Hecke correlations directly.

None of these has been tested at GPU frontier scale. We leave their investigation to future work.

8.2 Reproducibility

All findings reproduce from the `isomorphic-engine` Rust crate plus ≈ 30 CuPy / scipy Python scripts bundled with this manuscript. The canonical HP test battery is `DSC-2-engine/quantum_graphs/hilber` (50 phase-randomized surrogates, nfft=512). Riemann zero data from the Odlyzko ζ-zeros cache. Primitive Dirichlet L-function zeros computed by Hardy-Z sign changes.

Acknowledgements

We thank the authors of the Daugherty–Ward–Ryan `u24-spectral-operator` program for their open source code and the canonical HP test battery implementation, without which this survey would not have been possible. The Reeds J coupling matrix, x55 operator parameters, and Hardy-Z zero-finding code are all from OriginNeuralAI's public repositories.

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